

GRAHAM POSITIVITY OF QUANTUM DOUBLE SCHUBERT POLYNOMIALS

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ABSTRACT. We prove a positivity property for coefficients in the product expansion of quantum double Schubert polynomials.

1. INTRODUCTION

Let $S_\infty = \bigcup_{m \geq 1} S_m$, $\mathbf{x} = (x_1, x_2, \dots)$, $\mathbf{y} = (y_1, y_2, \dots)$, $\mathbf{t} = (t_1, t_2, \dots)$ and $\mathbf{q} = (q_1, q_2, \dots)$. Since $\mathbb{Z}[\mathbf{x}, \mathbf{t}][\mathbf{q}] = \bigoplus_{w \in S_\infty} \mathbb{Z}[\mathbf{t}][\mathbf{q}] \mathfrak{S}_w^q(\mathbf{x}; \mathbf{t})$, we can consider the following product expansion by extending scalars from $\mathbb{Z}[\mathbf{t}][\mathbf{q}]$ to $\mathbb{Z}[\mathbf{y}, \mathbf{t}][\mathbf{q}]$:

$$(1) \quad \mathfrak{S}_u^q(\mathbf{x}; \mathbf{y}) \mathfrak{S}_v^q(\mathbf{x}; \mathbf{t}) = \sum_{w \in S_\infty} C_{u,v}^w(\mathbf{y}, \mathbf{t}; \mathbf{q}) \mathfrak{S}_w^q(\mathbf{x}; \mathbf{t}).$$

Claim. For any $u, v, w \in S_\infty$,

$$C_{u,v}^w(\mathbf{y}, \mathbf{t}; \mathbf{q}) \in \mathbb{N}[t_i - y_j, q_k \mid i, j, k \geq 1].$$

For fixed $u, v \in S_\infty$, since the direct sum decomposition should be finite, we can choose n sufficiently large s.t. u, v and every permutation occurring in the expansion belong to S_n , while every variable occurring in its coefficients is among $y_1, \dots, y_n$, $t_1, \dots, t_n$, and $q_1, \dots, q_{2n-1}$. Write $\mathbf{x} = (x_1, \dots, x_{2n})$ and $\mathbf{q} = (q_1, \dots, q_{2n-1})$. Following the parameter arrangement method of Gao–Xiong [2], set $\mathbf{a} = (t_1, \dots, t_n, y_1, \dots, y_n)$, and let $\tau \in S_{2n}$ be the permutation s.t. $\tau \mathbf{a} = (y_1, \dots, y_n, t_1, \dots, t_n)$. (1) can be rewritten as:

$$(2) \quad \mathfrak{S}_u^q(\mathbf{x}; \tau \mathbf{a}) \mathfrak{S}_v^q(\mathbf{x}; \mathbf{a}) = \sum_{w \in S_{2n}} C_{u,v}^{w,\tau}(\mathbf{a}; \mathbf{q}) \mathfrak{S}_w^q(\mathbf{x}; \mathbf{a}).$$

Following Lam–Shimozono [4, Section 2.1], let e_i and $E_i^{(2n)}$ be the elementary symmetric polynomials and their quantum analogues, and write $J_{2n}^a = \langle e_i(\mathbf{x}) - e_i(\mathbf{a}) \rangle$ and $J_{2n}^{qa} = \langle E_i^{(2n)}(\mathbf{x}, \mathbf{q}) - e_i(\mathbf{a}) \rangle$. Follow Fomin–Gelfand–Postnikov [1, Theorem 1.1, Proposition 3.3] and extend the map by $\mathbb{Z}[\mathbf{a}]$ -linearity to a $\mathbb{Z}[\mathbf{q}, \mathbf{a}]$ -module automorphism of $\mathbb{Z}[\mathbf{x}, \mathbf{q}, \mathbf{a}]$, as in Lam–Shimozono [4, Section 3.3]. This induces a map $\bar{\theta} : \mathbb{Z}[\mathbf{x}, \mathbf{a}]/J_{2n}^a \rightarrow \mathbb{Z}[\mathbf{x}, \mathbf{a}, \mathbf{q}]/J_{2n}^{qa}$.

Classical equivariant Borel presentation gives an isomorphism between $\mathbb{Z}[\mathbf{x}, \mathbf{a}]/J_{2n}^a$ and $H_T^*(\text{Fl}_{2n})$; Kim’s presentation [3, Theorem 2] gives an isomorphism between $\mathbb{Z}[\mathbf{x}, \mathbf{a}, \mathbf{q}]/J_{2n}^{qa}$ and $QH_T^*(\text{Fl}_{2n})$. $\mathbb{Z}[\mathbf{x}, \mathbf{a}, \mathbf{q}]/J_{2n}^{qa} \cong QH_T^*(\text{Fl}_{2n})$; Lam–Shimozono [4, Theorem 5] identifies the basis $[\mathfrak{S}_w^q(\mathbf{x}; \mathbf{a})]_{J_{2n}^{qa}} \longleftrightarrow \sigma_w^T$. ϕ is the canonical $\mathbb{Z}[\mathbf{a}]$ -linear embedding sending $[X_w]^T$

to σ_w^T . We have the following commutative diagram:

$$\begin{array}{ccccc}
\mathbb{Z}[\mathbf{x}, \mathbf{a}] / J_{2n}^a & \xrightarrow[\cong]{\rho_{2n}^a} & H_T^*(\mathrm{Fl}_{2n}) & [\mathfrak{S}_w(\mathbf{x}; \mathbf{a})]_{J_{2n}^a} & \mapsto [X_w]^T & [\mathfrak{S}_u(\mathbf{x}; \tau \mathbf{a})]_{J_{2n}^a} & \xrightarrow{\text{red}} [\tau X_u]^T \\
\downarrow \bar{\theta} & \circlearrowleft & \downarrow \phi & \downarrow & \downarrow & \downarrow \text{blue} & \downarrow \text{green} \\
\mathbb{Z}[\mathbf{x}, \mathbf{a}, \mathbf{q}] / J_{2n}^{qa} & \xrightarrow[\cong]{\rho_{2n}^{qa}} & QH_T^*(\mathrm{Fl}_{2n}) & [\mathfrak{S}_w^q(\mathbf{x}; \mathbf{a})]_{J_{2n}^{qa}} & \mapsto \sigma_w^T & [\mathfrak{S}_u^q(\mathbf{x}; \tau \mathbf{a})]_{J_{2n}^{qa}} & \mapsto \phi([\tau X_u]^T)
\end{array}$$

Remark. The **red arrow** is what Gao–Xiong used in geometric proof [2]; the **blue arrow** is because of the $\mathbb{Z}[\mathbf{a}]$ -linearity of $\bar{\theta}$; the **green arrow** records that the twisted quantum Schubert class is $\phi([\tau X_u]^T)$, which allows us to apply Mihalcea’s EQLR.

Thus we can convert (2) to a geometric form in the flag variety. Decompose $\mathfrak{S}_u^q(\mathbf{x}; \tau \mathbf{a})$ along the quantum Schubert basis, embed the whole polynomial equation to the polynomial classes and map it into $QH_T^*(\mathrm{Fl}_{2n})$, then compare the coefficients of σ_w^T with the definition of EQLR in Mihalcea [5, Section 5]. Following Mihalcea’s notations [5], we have

$$(3) \quad C_{u,v}^{w,\tau}(\mathbf{a}; \mathbf{q}) = \sum_{\mathbf{d} \geq 0} \mathbf{q}^{\mathbf{d}} (\pi_{\mathcal{M}_{\mathbf{d}}})_*^T (\mathrm{ev}_1^*[\tau X_u]^T \cdot \mathrm{ev}_2^*[X_v]^T \cdot \mathrm{ev}_3^*[Y_w]^T \cap [\mathcal{M}_{\mathbf{d}}]^T).$$

2. PROOF OF CLAIM

Let $G = \mathrm{GL}_{2n}$, $B \subset G$ be a Borel subgroup, B^- be the opposite Borel, and $T = B \cap B^-$, so that $X = G/B = \mathrm{Fl}_{2n}$. Following Gao–Xiong [2], let N^- be the unipotent radical of B^- , $w_0^{(n)}$ be the longest element of S_n , and write $N^-(w) = N^- \cap wN^-w^{-1}$, $B^-(w) = TN^-(w)$. For $w \in S_{2n}$ we take the B^- -Schubert varieties $X_w = \overline{B^-wB}/B$ and the opposite Schubert varieties $Y_w = \overline{BwB}/B$. For a fixed effective degree $\mathbf{d}$, define $I_{u,v}^{\mathbf{d}} = (\mathrm{ev}_1, \mathrm{ev}_2)^{-1}(\tau X_u \times X_v)$ and $\Gamma_{u,v}^{\mathbf{d}} = (\mathrm{ev}_3)_*[I_{u,v}^{\mathbf{d}}]$.

Proposition 2.1. *Write $\mathcal{M}_{\mathbf{d}} = \overline{M}_{0,3}(X, \mathbf{d})$ with evaluation maps ev_i and $F = (\mathrm{ev}_1, \mathrm{ev}_2) : \mathcal{M}_{\mathbf{d}} \rightarrow X \times X$. For every $u, v \in S_n$, the incidence scheme $I_{u,v}^{\mathbf{d}} = F^{-1}(\tau X_u \times X_v)$ is reduced and locally irreducible, and each of its irreducible components has codimension $\ell(u) + \ell(v)$. Moreover,*

$$(4) \quad (\mathrm{ev}_1^*[\tau X_u]^T \cdot \mathrm{ev}_2^*[X_v]^T) \cap [\mathcal{M}_{\mathbf{d}}]^T = [I_{u,v}^{\mathbf{d}}]^T \quad \text{in } H_*^T(\mathcal{M}_{\mathbf{d}}).$$

Proof. Clearly, F is diagonally G -equivariant. Let $a_0 = 1 \times w_0^{(n)} \in S_{2n}$. Since $u, v \in S_n$, one has $a_0 X_u = X_u$, $a_0 X_v = X_v$, and $a_0^{-1} \tau a_0 = w_0^{(2n)}$; thus diagonal left translation by a_0^{-1} identifies $\tau X_u \times X_v$ with $w_0^{(2n)} X_u \times X_v$. The moduli space $\mathcal{M}_{\mathbf{d}}$ is irreducible by Thomsen’s theorem [6, Theorem 1]. By Mihalcea’s incidence lemma [5, Lemma 6.3], $I_{u,v}^{\mathbf{d}}$ is reduced, locally irreducible, and of codimension

$$\mathrm{codim}_X(w_0^{(2n)} X_u) + \ell(v) = \ell(u) + \ell(v).$$

In particular, writing $W = \tau X_u \times X_v \subset X \times X$, every irreducible component of $F^{-1}(W) = I_{u,v}^{\mathbf{d}}$ has codimension $\mathrm{codim}_{X \times X} W = \ell(u) + \ell(v)$ and algebraic multiplicity one. Mihalcea’s

formula (2'') [5, Section 3.1], applied to $F : \mathcal{M}_{\mathbf{d}} \rightarrow X \times X$ (with $X \times X$ smooth), therefore yields

$$F^*[W]^T \cap [\mathcal{M}_{\mathbf{d}}]^T = [I_{u,v}^{\mathbf{d}}]^T \quad \text{in } H_*^T(\mathcal{M}_{\mathbf{d}}).$$

Since $F^*[W]^T = \text{ev}_1^*[\tau X_u]^T \cdot \text{ev}_2^*[X_v]^T$, this is precisely (4). $\square$

Lemma 2.2. *The cycle $\Gamma_{u,v}^{\mathbf{d}}$ is an effective $B^-(\tau)$ -invariant cycle on X .*

Proof. The variety X_v is B^- -invariant. If $h = \tau h' \tau^{-1} \in N^-(\tau) = N^- \cap \tau N^- \tau^{-1}$, then $h(\tau X_u) = \tau h' X_u \subseteq \tau X_u$. Hence $\tau X_u \times X_v$ is $B^-(\tau)$ -invariant, and so is its preimage $I_{u,v}^{\mathbf{d}} = F^{-1}(\tau X_u \times X_v)$. By Proposition 2.1, $I_{u,v}^{\mathbf{d}}$ is a reduced closed subscheme of $\mathcal{M}_{\mathbf{d}}$, so its fundamental cycle is effective. Since ev_3 is proper, the push-forward $\Gamma_{u,v}^{\mathbf{d}} = (\text{ev}_3)_*[I_{u,v}^{\mathbf{d}}]$ is an effective cycle on X . Moreover ev_3 is G -equivariant, so $\Gamma_{u,v}^{\mathbf{d}}$ is $B^-(\tau)$ -invariant. $\square$

Theorem 2.3. *With $\Gamma_{u,v}^{\mathbf{d}} = (\text{ev}_3)_*[I_{u,v}^{\mathbf{d}}]$, one has:*

(i) (Push-forward; [5, Section 3.1, formula (2)], applied with $f = \text{ev}_3$, $g = F$, and $V = \tau X_u \times X_v$.)

$$(5) \quad [\Gamma_{u,v}^{\mathbf{d}}]^T = (\text{ev}_3)_*^T (\text{ev}_1^*[\tau X_u]^T \cdot \text{ev}_2^*[X_v]^T) \quad \text{in } H_T^*(X).$$

(ii) (Projection; [5, Section 3.1, formula (1)], applied with $f = \text{ev}_3$, $a = [Y_w]^T$, and $b = \text{ev}_1^*[\tau X_u]^T \cdot \text{ev}_2^*[X_v]^T$.)

$$(6) \quad (\pi_{\mathcal{M}_{\mathbf{d}}})_*^T (\text{ev}_1^*[\tau X_u]^T \cdot \text{ev}_2^*[X_v]^T \cdot \text{ev}_3^*[Y_w]^T \cap [\mathcal{M}_{\mathbf{d}}]^T) = (\pi_X)_*^T ([\Gamma_{u,v}^{\mathbf{d}}]^T [Y_w]^T).$$

Proof. For (i), Proposition 2.1 and Lemma 2.2 supply the conditions of [5, Section 3.1, formula (2)] (expected codimension, multiplicity one, and T -invariant components), and the resulting push-forward is $[\Gamma_{u,v}^{\mathbf{d}}]^T$ by definition of $\Gamma_{u,v}^{\mathbf{d}}$.

For (ii), write $b = \text{ev}_1^*[\tau X_u]^T \cdot \text{ev}_2^*[X_v]^T$. Formula (1) of [5, Section 3.1] gives $(\text{ev}_3)_*^T (\text{ev}_3^*[Y_w]^T \cdot b) = [Y_w]^T \cdot (\text{ev}_3)_*^T(b)$. Using (i) and $\pi_{\mathcal{M}_{\mathbf{d}}} = \pi_X \circ \text{ev}_3$, this yields (6). $\square$

Theorem 2.4. *There is a unique expansion*

$$(7) \quad [\Gamma_{u,v}^{\mathbf{d}}]^T = \sum_{z \in S_{2n}} p_{u,v}^{z,\mathbf{d}}(\mathbf{a}) [X_z]^T \quad \text{in } H_T^*(X),$$

and

$$p_{u,v}^{z,\mathbf{d}}(\mathbf{a}) \in \mathbb{N}[-\alpha]_{\alpha \in I(\tau)} = \mathbb{N}[t_i - y_j : 1 \leq i, j \leq n].$$

Here $I(\tau) = \{y_j - t_i : 1 \leq i, j \leq n\}$ is the left inversion set of τ , in the notation of Gao-Xiong [2, Sections 2.1 and 2.3].

Proof. The classes $\{[X_z]^T : z \in S_{2n}\}$ form a $\mathbb{Z}[\mathbf{a}]$ -basis of $H_T^*(X)$, so the expansion exists and is unique. Since $\Gamma_{u,v}^{\mathbf{d}}$ is $B^-(\tau)$ -invariant by Lemma 2.2, apply [2, Theorem 2.3 and Corollary 2.4] then we get the desired expansion. $\square$

Combining (3), (6) and (7), we obtain

$$C_{u,v}^{w,\tau}(\mathbf{a}; \mathbf{q}) = \sum_{\mathbf{d} \geq 0} \mathbf{q}^{\mathbf{d}} (\pi_X)_* ([\Gamma_{u,v}^{\mathbf{d}}]^T [Y_w]^T) = \sum_{\substack{\mathbf{d} \geq 0 \\ z \in \bar{S}_{2n}}} p_{u,v}^{z,\mathbf{d}}(\mathbf{a}) \mathbf{q}^{\mathbf{d}} (\pi_X)_* ([X_z]^T [Y_w]^T) = \sum_{\substack{\mathbf{d} \geq 0 \\ z \in \bar{S}_{2n}}} p_{u,v}^{z,\mathbf{d}}(\mathbf{a}) \mathbf{q}^{\mathbf{d}} \delta_{z,w}.$$

Apply Theorem 2.4 then we have $C_{u,v}^{w,\tau}(\mathbf{a}; \mathbf{q}) \in \mathbb{N}[t_i - y_j, q_k \mid i, j, k \geq 1]$. This proves the claim.

REFERENCES

1. Sergey Fomin, Sergei Gelfand, and Alexander Postnikov, *Quantum Schubert polynomials*, Journal of the American Mathematical Society **10** (1997), no. 3, 565–596.
2. Yibo Gao and Rui Xiong, *Graham positivity of triple Schubert calculus*, 2025, arXiv:2506.09421 [math.CO].
3. Bumsig Kim, *On equivariant quantum cohomology*, International Mathematics Research Notices (1996), no. 17, 841–851.
4. Thomas Lam and Mark Shimozono, *Quantum double Schubert polynomials represent Schubert classes*, Proceedings of the American Mathematical Society **142** (2014), no. 3, 835–850.
5. Leonardo Constantin Mihalcea, *Positivity in equivariant quantum Schubert calculus*, American Journal of Mathematics **128** (2006), no. 3, 787–803.
6. Jesper Funch Thomsen, *Irreducibility of $\overline{M}_{0,n}(g/p, \beta)$* , International Journal of Mathematics **9** (1998), no. 3, 367–376.